

Thesis Section Build Report

Output folder: D:\TUBAF\Master_Thesis\Abaqus_trial\runs\chaboche_umat\thesis_cycle_jump_section

Copied files

- tables\chaboche_vp_v1_amplitude_sweep_summary.csv
- tables\chaboche_vp_v1_cyclic_eps005_10cycles_summary.csv
- tables\chaboche_vp_v1_cyclic_eps005_10cycles_cycle_increments.csv
- tables\chaboche_cycle_jump_predictions.csv
- tables\chaboche_cycle_jump_curve_1_to_1000.csv
- tables\chaboche_vp_v1_cyclic_eps005_20cycles_summary.csv
- tables\chaboche_vp_v1_cyclic_eps005_20cycles_cycle_increments.csv
- figures\source_chaboche_vp_v1_amplitude_sweep_stress_strain.svg
- figures\source_chaboche_vp_v1_cyclic_eps005_10cycles_selected_loops.svg
- figures\source_chaboche_eps005_10cycles_delta_sdv1_per_cycle.svg
- figures\source_chaboche_cycle_jump_sdv1_prediction.svg
- figures\source_chaboche_cycle_jump_vs_explicit_20cycles.svg

Generated figures

- figures/fig00_geometry_material_model.svg
- figures/fig00_geometry_material_model.png
- figures/fig01_amplitude_sweep_stress_strain.svg
- figures/fig01_amplitude_sweep_stress_strain.png
- figures/fig02_10cycle_selected_hysteresis_loops.svg
- figures/fig02_10cycle_selected_hysteresis_loops.png
- figures/fig03_delta_sdv1_per_cycle.svg
- figures/fig03_delta_sdv1_per_cycle.png
- figures/fig04_cycle_jump_sdv1_prediction.svg
- figures/fig04_cycle_jump_sdv1_prediction.png
- figures/fig05_cycle_jump_vs_explicit_20cycles.svg
- figures/fig05_cycle_jump_vs_explicit_20cycles.png

Additional model figure:

- figures/fig00_geometry_material_model.svg/png shows the single C3D8 block geometry, displacement boundary conditions, and Chaboche-v1 material/state-variable workflow.
- scripts/make_geometry_material_model_figure.py regenerates the geometry/material-model figure.
- figures/fig00b_actual_abaqus_geometry.svg/png is reconstructed directly from the Abaqus input deck node coordinates and C3D8 element connectivity.
- scripts/make_actual_abaqus_geometry_figure.py regenerates the actual Abaqus geometry figure from chaboche_vp_v1_cyclic_eps005_20cycles.inp.

Generated tables

- tables/table01_amplitude_sweep_summary.csv
- tables/table02_cycle_jump_validation_summary.csv

LaTeX files

- latex/cycle_jump_chaboche_section.tex
- latex/cycle_jump_chaboche_figures_only.tex

Missing source files

- None

PNG conversion

- PNG generation succeeded: yes

Final validation result

- Predicted SDV1 at cycle 20: 0.1421214351
- Explicit SDV1 at cycle 20: 0.1420256943
- Relative error: 0.0674%

Material-model scope

The Chaboche unified viscoplastic UMAT has been successfully implemented and validated at the computational workflow level. It reproduces nonzero stress, reaction force, viscoplastic strain accumulation, cyclic hysteresis, and a stable per-cycle internal-variable increment suitable for cycle-jump prediction. The current Chaboche-v1 parameter set is a demonstration/validation set, not yet a fully calibrated 316 stainless steel material model.

Standalone PDF Compilation

- Wrapper file: `cycle_jump_chaboche_standalone.tex`
- PDF file: `cycle_jump_chaboche_standalone.pdf`
- Compile command: `latexmk -pdf -interaction=nonstopmode -halt-on-error .\cycle_jump_chaboche_standalone.tex`
- Compile status: succeeded
- PDF pages after adding actual Abaqus geometry figure: 6
- PDF size after adding actual Abaqus geometry figure: 964447 bytes
- Note: MiKTeX printed a maintenance warning that user/administrator updates are out-of-sync, but the PDF was generated successfully.

Level-2 Cycle-Jump Preparation Integration

Integration Date: May 8, 2026

The Level-2 preparation subsection was added to the standalone thesis package:

- New LaTeX file: `latex/chaboche_level2_cycle_jump_preparation_section.tex` (includable section format)
- Original Level-2 document: `../CHABOCHE_V1_LEVEL2_CYCLE_JUMP_PREPARATION_SUMMARY.md`
- Standalone wrapper updated: `cycle_jump_chaboche_standalone.tex` now includes both Level-1 scalar SDV1 section and Level-2 full-STATEV preparation section

Content added: - STATEV inventory table (15 state variables with symbols, meanings, and categories) - Full-state cycle-history extraction findings and stability classification - Vector-valued STATEV cycle-jump control analysis (original $\Delta N=2$, target cycle 12) - Exact-output diagnostic branch findings (increment-schedule sensitivity, 5.12% cycle-20 difference) - Decision rationale for deferring STATEV injection to Level-3 - Method level distinctions (Level 1: postprocessing only, Level 2: preparation diagnostics, Level 3: full integration)

Updated PDF: - New page count: 8 pages (previously 6) - New size: 1070847 bytes - Compilation status: successful with booktabs package support

Thesis narrative now covers: 1. Validated scalar SDV1 cycle-jump demonstration (Level 1) 2. Full-state preparation diagnostics identifying robustness requirements (Level 2) 3. Clear identification of why Level-3 injection is deferred pending UMAT robustness improvements

Stage 3 Increment-Schedule Sensitivity Integration

Integration Date: May 9, 2026

The Stage 3 increment-schedule sensitivity subsection was added to the standalone thesis package immediately after the Level-2 preparation section.

Added and copied into the thesis package:

- latex/chaboche_increment_sensitivity_section.tex
- tables/chaboche_increment_sensitivity_summary.csv
- tables/chaboche_eps005_20cycles_dt_original_output_statev_history.csv
- tables/chaboche_eps005_20cycles_dtmax_0p01_statev_history.csv
- tables/chaboche_eps005_20cycles_dtmax_0p005_inc6000_statev_history.csv
- figures/chaboche_increment_sensitivity_statev1_vs_dmax.svg
- figures/chaboche_increment_sensitivity_statev1_vs_cycle.svg

Content added: - Cycle-20 summary table for the three controlled DMAX cases - LaTeX/PGFPlots rendering of STATEV1 versus DMAX - LaTeX/PGFPlots rendering of STATEV1 versus cycle for the three completed cases - Thesis conclusion that increment-size sensitivity is confirmed and Level-3 injection remains deferred

Updated PDF: - New page count: 10 pages - New size: 1084855 bytes - Compilation status: successful

Thesis narrative now covers: 1. Validated scalar SDV1 cycle-jump demonstration (Level 1) 2. Full-state preparation diagnostics identifying robustness requirements (Level 2) 3. Increment-schedule sensitivity evidence showing the UMAT remains increment-size sensitive (Stage 3)

Predicted-State FE Cycle-Jump Continuation Integration

Integration Date: May 9, 2026

The final predicted-state FE cycle-jump subsection was added to the standalone thesis package after the Stage 3 increment-schedule sensitivity section.

Added and copied into the thesis package:

- latex/chaboche_predicted_fe_cycle_jump_section.tex
- tables/table_predicted_fe_cycle_jump_summary.csv
- tables/table_multitarget_jump_scan.csv

Updated:

- cycle_jump_chaboche_standalone.tex
- cycle_jump_chaboche_standalone.pdf

New section name:

- Predicted-State FE Cycle-Jump Continuation

Content added:

- Method paragraph covering cycle-space prediction, SDVINI STATEV injection, SIGINI residual-stress injection, one computed continuation cycle, and comparison against full no-skip Abaqus references.
- Stage 5B clean predicted FE cycle-jump result:
 - Route: cycle 10 -> predicted cycle 19 -> cycle 20
 - Skipped intermediate FE cycles: 8
 - STATEV1 relative error: 0.049427%
 - S11 relative error: 0.127013%
 - Decision: clean success
- Stage 6C multi-target prediction scan:
 - Target 29 -> 30: acceptable exploratory candidate
 - Target 39 -> 40: not headline candidate
 - Target 49 -> 50: not headline candidate
- Stage 6D larger predicted FE cycle-jump result:
 - Route: cycle 10 -> predicted cycle 29 -> cycle 30
 - Skipped intermediate FE cycles: 18
 - Computed route: 11 cycles instead of 30
 - Route reduction: 63.33%
 - STATEV1 relative error: 0.0458269%
 - S11 relative error: 2.34366%
 - Outcome: acceptable exploratory success

Limitation stated in the thesis text:

The remaining limitation is not Abaqus initialization, SDVINI, SIGINI, or UMAT continuation. These mechanisms were verified. The remaining limitation is the accuracy of first-order stress/backstress extrapolation for larger cycle jumps.

Updated PDF:

- New page count: 11 pages
- New size: 1091019 bytes
- Compilation status: successful
- Compile command: `latexmk -pdf -interaction=nonstopmode -halt-on-error .\cycle_jump_chaboche_standalone.`
- Note: MiKTeX again printed user/administrator update warnings, but the PDF was generated successfully.

Thesis narrative now covers:

1. Validated scalar SDV1 cycle-jump demonstration (Level 1)
2. Full-state preparation diagnostics identifying robustness requirements (Level 2)
3. Increment-schedule sensitivity evidence for the Chaboche-v1 UMAT (Stage 3)
4. Predicted-state FE cycle-jump continuation through SDVINI/SIGINI, including clean Stage 5B and larger exploratory Stage 6D validation

Stage 7B Adaptive Jump-Size Selection Integration

Integration Date: May 10, 2026

A short subsection was added after the predicted FE cycle-jump continuation results:

- Adaptive Jump-Size Selection Inspired by Nesnas--Saouni

Updated:

- `latex/chaboche_predicted_fe_cycle_jump_section.tex`
- `cycle_jump_chaboche_standalone.pdf`
- `THESIS_SECTION_BUILD_REPORT.md`

Content added:

- The paper damage variable D is not used directly because the present model is a Chaboche viscoplastic UMAT without damage.
- D is replaced by a generalized Chaboche cycle-control quantity Y_i .
- The paper-style jump budget is written as an admissible state change $A_i = \tau_i S_i$.
- The per-variable jump estimate is documented as $\Delta N_i = \text{floor}(\eta A_i / (\text{abs}(\text{mean } \Delta Y_i) + \epsilon))$.
- Accumulated viscoplastic strain p is treated as an accuracy monitor rather than the global restart limiter.
- Backstress, viscoplastic strain tensor, and residual stress consistency control the grouped restart recommendation.
- The grouped Stage 7B recommendation $\Delta N_{\text{restart}} = 17$ is stated as close to the validated Stage 6D exploratory FE jump with $\Delta N = 19$.

Updated PDF:

- New page count: 12 pages
- New size: 1105468 bytes
- Compilation status: successful
- Compile command: `latexmk -pdf -interaction=nonstopmode -halt-on-error .\cycle_jump_chaboche_standalone.`
- Note: MiKTeX printed user/administrator update warnings, but the PDF was generated successfully.

Thesis narrative now covers:

1. Validated scalar SDV1 cycle-jump demonstration (Level 1)
2. Full-state preparation diagnostics identifying robustness requirements (Level 2)
3. Increment-schedule sensitivity evidence for the Chaboche-v1 UMAT (Stage 3)
4. Predicted-state FE cycle-jump continuation through SDVINI/SIGINI, including clean Stage 5B and larger exploratory Stage 6D validation
5. Paper-inspired grouped adaptive ΔN selection with Stage 7B recommendation $\Delta N_{\text{restart}} = 17$

Stage 7C Adaptive Target Abaqus Validation Integration

Integration Date: May 10, 2026

The Stage 7C direct Abaqus validation was added to the adaptive jump-size subsection.

Updated:

- latex/chaboche_predicted_fe_cycle_jump_section.tex
- cycle_jump_chaboche_standalone.pdf
- THESIS_SECTION_BUILD_REPORT.md

Content added:

- Stage 7B formula-selected recommendation: $\Delta N_{\text{restart}} = 17$.
- Direct validation route: cycle 10 -> predicted cycle 27 -> cycle 28.
- Skipped intermediate FE cycles: 16.
- First-frame injection check:
 - STATEV1 absolute error: $1.39693476231\text{e-}09$
 - S11 absolute error: $3.39113159953\text{e-}06$ MPa
- Final cycle-28 comparison against the interpolated 50-cycle Abaqus reference:
 - STATEV1 relative error: 0.0231584782019%
 - S11 relative error: 2.36494669088%
- Stage 7C outcome: `accepted_exploratory_success`.

Thesis interpretation:

Stage 7C confirms that the grouped adaptive controller is not only close to the manually validated Stage 6D jump, but also produces an Abaqus-validated formula-selected exploratory cycle jump.

Updated PDF:

- New page count: 12 pages
- New size: 1106447 bytes
- Compilation status: successful
- Compile command: `latexmk -pdf -interaction=nonstopmode -halt-on-error .\cycle_jump_chaboche_standalone.`
- Note: MiKTeX printed user/administrator update warnings, but the PDF was generated successfully.

Stage 8 Final Thesis Polishing Snapshot

Integration Date: May 10, 2026

The cycle-jump thesis package was consolidated through the adaptive Abaqus validation stage.

Updated:

- latex/chaboche_predicted_fe_cycle_jump_section.tex
- cycle_jump_chaboche_standalone.pdf
- THESIS_SECTION_BUILD_REPORT.md

Content added:

- Final summary table comparing Stage 5B, Stage 6D, Stage 7B, and Stage 7C:
 - Stage 5B: clean predicted FE jump, $\Delta N = 9$, 8 skipped cycles, clean success
 - Stage 6D: larger manually selected jump, $\Delta N = 19$, 18 skipped cycles, accepted exploratory success
 - Stage 7B: grouped adaptive controller, $\Delta N_{\text{restart}} = 17$, recommendation stage
 - Stage 7C: formula-selected FE validation, $\Delta N = 17$, 16 skipped cycles, accepted exploratory success
- Limitations and future-work sentence covering material calibration, improved stress/backstress extrapolation, multi-element tests, and longer cyclic histories.

Reference/order check:

- Figure numbering in `.aux`: 1 through 9, in order.
- Table numbering in `.aux`: 1 through 6, in order.
- Final summary table label: `tab:final_cycle_jump_summary`.

- No missing-reference or citation warnings were reported by the LaTeX log.
- Remaining log issues are pre-existing overfull hbox warnings.

Updated PDF:

- New page count: 12 pages
- New size: 1107407 bytes
- Compilation status: successful
- Compile command: `latexmk -pdf -interaction=nonstopmode -halt-on-error .\cycle_jump_chaboche_standalone.`
- Note: MiKTeX printed user/administrator update warnings, but the PDF was generated successfully.

Current thesis-package status:

The cycle-jump thesis section is complete through the final Stage 8 polishing snapshot. It now documents the clean small jump, larger exploratory jump, grouped adaptive controller, direct adaptive-target Abaqus validation, and remaining limitations/future work.

Supervisor Note: Nesnas–Saanouni Comparison

Integration Date: May 10, 2026

A short thesis-safe scope paragraph was added to the adaptive jump-size subsection to clarify that the present work is inspired by the Nesnas–Saanouni damage-based cycle-jump strategy, but is not a direct implementation of their coupled damage-viscoplasticity formulation. The paragraph states that the Chaboche UMAT does not include an explicit damage variable, so the damage-control quantity is replaced by generalized Chaboche state and stress-control variables.

Created:

- `NESNAS_SAAOUNI_COMPARISON_SUPERVISOR_NOTE.md`

This supervisor/discussion note keeps the full comparison table outside the thesis main text under the title `Comparison Between Nesnas--Saanouni Damage-Based Cycle Jumping and the Present Chaboche UMAT Cycle-Jump Workflow`.

Updated PDF:

- Page count: 12 pages
- PDF size: 1107982 bytes
- Compilation status: successful
- Compile command: `latexmk -pdf -interaction=nonstopmode -halt-on-error .\cycle_jump_chaboche_standalone.`
- Note: MiKTeX printed the same user/administrator update warning; the PDF was generated successfully.

PDF copies were also generated for review: `NESNAS_SAAOUNI_COMPARISON_SUPERVISOR_NOTE.pdf` and `THESIS_SECTION_BUILD_REPORT.pdf`.